

Scene-First Vehicle-Evidenced Fusion for Urban Flood-Depth Estimation from Street Images

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Abstract—Street-level flood-depth awareness is important for road safety, route planning, and emergency response, yet conventional flood monitoring systems often lack fine-grained information at the scale of individual urban roads. This paper studies urban flood-depth estimation from street images using vehicles as practical local waterline references. Specifically, we proposed a scene-first vehicle-evidenced fusion model that treats the global scene as the anchor, selects the most flood-informative vehicle in the scene, refines local waterline evidence with attention, and applies bounded upward or downward ordinal corrections through a directional reliability-gated correction module. To facilitate this study, a paired global-local image dataset was constructed, where each flood scene is linked to cropped vehicle images. Experimental results show that cropped vehicles are most effective when treated as conditional evidence rather than as direct substitutes for scene-level reasoning.

Index Terms—Urban flooding, flood-depth estimation, ordinal classification, vehicle-guided fusion, directional reliability-gated correction

I. INTRODUCTION

Urban flooding disrupts transportation networks, emergency response, and daily mobility, directly threatening public safety and economic activity [1]. For road users and emergency managers, the key question is not only whether a street is flooded, but how severe the local water level is. A shallow puddle, water near a wheel hub, or water above a vehicle fender imply different levels of risk [2]. Therefore, a timely awareness of the depth of floods at the street-level can support route choice, emergency prioritization, and urban public-safety management.

Conventional flood monitoring provides essential regional information, but is not always sufficient for real-time road-scale depth awareness. Remote sensing offers broad coverage, while is limited by revisit time, weather, cloud cover, and spatial resolution in dense urban areas [3]. Water-level gauges and fixed sensors provide direct measurements, but are spatially sparse and are often installed near selected rivers, canals, coastal zones, or drainage assets rather than on ordinary streets [4]. Hydrodynamic models support planning and scenario analysis, but street-level simulation depends on topography, drainage data, and parameter assumptions [5]. Ground-level visual observations can complement these systems rather than replace them.

Street images from mobile phones, social networks, traffic cameras, dash cameras, and other public channels can provide

timely visual evidence of local flood conditions [6]–[8]. They capture water extent, road surfaces, vehicles, pedestrians, buildings, reflections, and other local reference cues, making them a useful complementary source for location-specific flood awareness.

Prior image-based flood-depth studies generally follow two complementary directions. First, reference objects such as people, traffic signs, poles, vehicles, buses, or gauges are detected, and water depth is inferred from their visible submergence [9], [10]. Second, visual learning models use detection, segmentation, attention-enhanced classifiers, or end-to-end deep networks to estimate water depth or waterlogging severity from flood images [11]–[13]. However, models learned from full images may miss fine local waterline evidence, while object-based methods often assume that the selected reference object is located in the flood-relevant area and that its local waterline represents the whole scene.

Real urban flood images make these assumptions more unreliable. A detected vehicle may be on an elevated dry road while deeper flooding appears in the background; it may also be beside a canal or river rather than on a flooded roadway. However, a vehicle can provide strong waterline evidence when the global scene underestimates flood depth. The model should therefore reason about the relation between the global scene, vehicle evidence, and the flood-relevant region. Local evidence should be incorporated into the overall prediction according to its reliability and consistency in the scene.

Motivated by this, we proposed a *scene-first vehicle-evidenced (SF-VE) fusion framework*, where the full image serves as the global scene anchor, while the detected vehicles provide conditional evidence to correct the prediction from the global scene. As illustrated in Fig. 1, the model has a dual-branch architecture with a global scene branch and a local vehicle-evidence branch. The information extracted from two branches is then fused through a Vehicle-Evidence Selection and Refinement Module (VE-SRM) and a Directional Reliability-Gated Correction Module (DRG-CM). The VE-SRM selects the vehicle crop with the highest expected depth score, computed from the local-level ordinal probability distribution, and then refines the selected vehicle feature using global scene context. The DRG-CM employs the refined local vehicle features to improve the prediction from the global scene branch, where the upward path corrects the

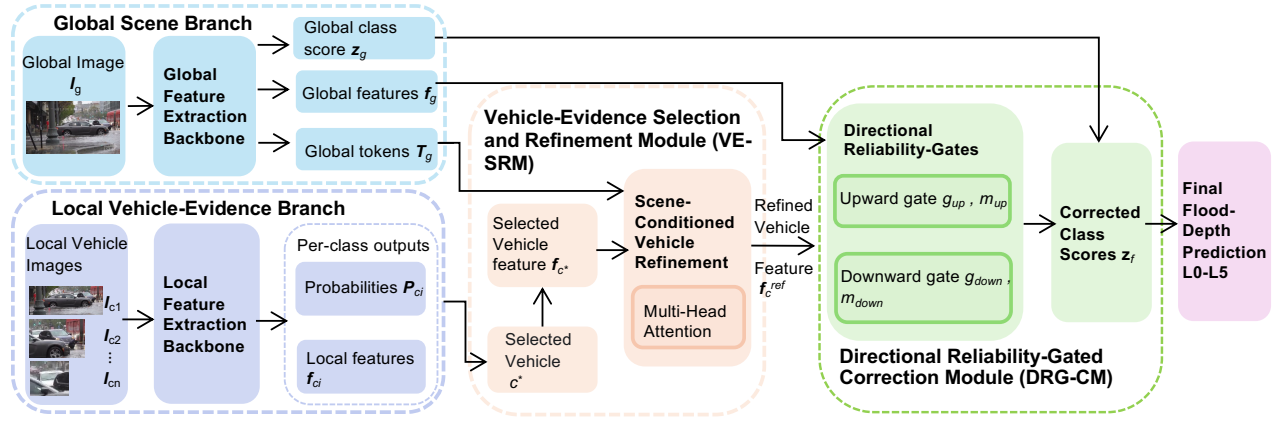


Fig. 1. An overview of the proposed scene-first vehicle-evidenced fusion model with a dual-branch architecture. The global scene serves as the anchor to select and refine the most flood-informative vehicle evidence through the VE-SRM. Then, the DRG-CM employs the refined local vehicle evidence to produce bounded upward or downward ordinal corrections to the prediction from the global scene.

underestimation when the selected vehicle evidence shows a reliable deeper waterline, while the downward path corrects overestimation from misleading water regions, reflections, or non-road water bodies.

Our main contributions are summarized as follows:

- A scene-first vehicle-evidenced fusion model with two novel modules was proposed, which employs the global scene as the anchor, selects and refines the vehicle-based evidence via the VE-SRM, and corrects the global-scene-based prediction through the DRG-CM.
- A paired global-local urban flood-depth (GL-UFD) dataset was constructed to facilitate this study, with flood depth labels for the global scene and each detected vehicle.

Experimental results on the GL-UFD dataset show that the proposed SF-VE fusion framework outperforms baselines using only the global scene or the global-local feature-level fusion, demonstrating the effectiveness of employing local vehicles as conditional evidence.

II. RELATED WORK

A. Scene-Level Image-Based Flood-Depth Estimation

Early image-based flood-depth estimation often relied on calibrated or fixed-scenario visual monitoring. Fixed cameras near rivers, canals, gauges, or flood-prone road segments can support continuous measurement when camera pose, scene geometry, and reference objects are known [14]. However, these mappings are usually tied to site-specific viewpoints, backgrounds, and calibration settings. They also require camera deployment, calibration, and maintenance, which limit their scalability for broad street-level flood awareness.

To improve scalability, recent work has used uncontrolled images from social networks, street photos, mobile devices, traffic cameras, and other public sources. Chaudhary et al. [6] estimated flood-water level from social media images, showing that online images can provide useful local flood evidence during flood events. Alizadeh et al. [15] combined image

processing and deep neural networks for flood-depth mapping in street-level photos. Li et al. [16] studied the automatic detection of urban flood-water depth from social media images, further demonstrating the potential of public images for near-real-time flood assessment. Other methods formulated the problem as image classification using improved CNNs, attention modules, or residual networks. For example, Wu et al. [8] used a CBAM-improved ResNet50 for urban water-logging depth-level identification; and Du et al. [2] proposed an improved ResNet-based method for water-depth estimation from social media images.

These scene-level methods are important because the full image contains road layout, water extent, reflections, buildings, pedestrians, and surrounding context. However, these methods may miss fine waterline cues around small, partially occluded, or background reference objects. This limitation motivates the use of local reference evidence, but only when the evidence is consistent with the flood-relevant scene.

B. Flood-Level Estimation Using Reference Objects

Reference-object reasoning is common because uncontrolled images usually lack metric scale. Human-centered methods estimate flood severity by comparing visible waterlines with body landmarks such as ankle, knee, hip, or chest [3], [16]. These rules are intuitive, but human-based estimates are sensitive to body-size variation, age, pose, clothing, and incomplete visibility of key body landmarks. Anthropometric standards and national statistics further show that body measurements vary across populations, which weakens fixed body-landmark assumptions for consistent flood-depth labeling [17], [18].

Other studies use infrastructure or street objects, including traffic signs, poles, curbs, roadside barriers, water gauges, and building features [6], [15]. These references provide useful scale cues, but are not consistently visible. Traffic signs depend on roadway context and placement standards [19]. Curbs help mainly in shallow water but disappear in deeper water. Gauges and markers are often installed near rivers, canals, reservoirs,

or coastal areas, which may not represent ordinary urban roadway flooding.

Vehicle-based flood estimation is especially relevant to urban mobility because vehicles are common in road scenes and directly tied to travel safety. Prior work has estimated depth from flooded-vehicle geometry [7], detected flood levels with flooded vehicle datasets [9], and explored bus-based flood-level detection [1]. Nevertheless, trucks, buses, bicycles, and other vehicle categories vary widely in geometry and frequency of occurrence. In this work, we focus on car-class vehicles, especially sedans, SUVs, and pickups, because they are common in urban road images and provide visually interpretable wheel landmarks, including the wheel lower rim, hub, upper rim, and fender.

The key limitation of reference-object-based methods is representativeness. A visible vehicle may be on an elevated dry road, beside a canal, or outside the deepest flooded region; while a small local vehicle may provide the clearest available waterline evidence when the global image underestimates depth. Our work addresses this limitation through a scene-first vehicle-evidenced design: the full image remains the scene anchor, while local vehicles are treated as conditional evidence for bounded correction rather than as direct substitutes for the prediction from the global scene.

III. DATASET CONSTRUCTION

A. Flood Image Collection and Vehicle-Crop Generation

The pipeline for constructing the GL-UFD dataset is shown in Fig. 2. Candidate flood images were collected from public web image search using “flooding” combined with keywords such as “vehicle”, “urban”, “stop sign”, “traffic sign”, “gauge”, “nature”, “people”, “weather”, and “light”. The initial pool contained approximately 9,969 images. After removing duplicate, near-duplicate, irrelevant, and low-quality images through source/file inspection and manual review, YOLOv11 vehicle detection [20] was used to detect vehicles. We retained the car-class and excluded trucks and buses because the label rule relies on relatively consistent car wheel geometry/size. Global images without detected cars are removed so that each remaining scene has at least one associated local vehicle. For each detected car, the bounding box is expanded by 1.5 times around the box center before cropping to preserve the wheel, nearby road surface, and waterline context.

The final paired dataset contains 2,660 global images and 7,434 local vehicle images, as summarized in Table I. Both global images and local vehicle images are manually labeled: global images use L0–L5 labels, while local vehicle images use L0–L5 plus N for locally indeterminate cases. For training and evaluation purposes, we first split the global images into train, validation, and test sets; all local vehicle images of a global image inherit the same split, preventing leakage across partitions.

B. Vehicle-Referenced Ordinal Label Rule

Rather than assigning continuous depth, we define an ordinal L0–L5 rule using car-class vehicle wheel landmarks,

TABLE I
DATASET SPLIT SUMMARY.

Data type	Train	Val	Test	Total	Labels
Paired Global Images	1864	397	399	2660	L0–L5
Local Vehicle Images	5220	1091	1123	7434	L0–L5/N

as illustrated in Fig. 3. This reflects the uncertainty of uncontrolled images, where viewpoint, occlusion, reflection, and vehicle pose make exact depth annotation difficult [6]. Wheel landmarks are visually interpretable and can be cross-checked with aligned vehicle parts such as bumpers, door sills, lights, and fenders.

For local vehicle-based labeling, L0 indicates dry or wet road surfaces without standing water; these cases are merged because they are visually indistinguishable in uncontrolled images and have the same operational interpretation. L1 denotes shallow water up to roughly 4 inches near the lower wheel rim or curb, L2 denotes approximately 4–10 inches near the wheel hub or mid-wheel, L3 denotes approximately 10–18 inches near the upper wheel rim, L4 denotes approximately 18–28 inches near the fender, and L5 denotes above-fender water (over 28 inches). When the wheel is partially occluded, we use aligned cues such as the bottom of the door, door sill, bumper, lights, trunk lid, and exhaust pipe. The N label is assigned to vehicles that are too blurred, occluded, truncated, or ambiguous for local depth judgment to decide a flood level.

Global-image labels use the same L0–L5 scale, but are assigned at the scene level. Vehicles are the primary reference, but curbs, road surface, pedestrians, water extent, and background flooding are also considered because a visible vehicle may not lie in the deepest or most relevant flooded region. Therefore, the global label is not always equal to the maximum depth label of its associated vehicles. This protocol reflects the core task: vehicle evidence can be useful, but the global label should represent the flood-relevant scene rather than a single local vehicle.

IV. METHODOLOGY

A. An Overview of Scene-First Vehicle-Evidenced Fusion

Given a global image $\mathbf{I}_g$ and detected vehicle images $\{\mathbf{I}_{c,i}\}_{i=1}^n$, the goal is to predict the scene-level flood-depth label $y_g \in \{L_0, \dots, L_5\}$. As illustrated in Fig. 1, the SF-VE framework is scene-first, because a detected vehicle may be informative, irrelevant, or misleading depending on whether the vehicle is located in the flood-relevant region. Therefore, the full image provides the anchor prediction, while local vehicles provide bounded correction evidence.

Specifically, the SF-VE framework has a dual-branch architecture: the global scene branch and local vehicle-evidence branch using Swin-S backbones [21] pretrained on the global images and cropped vehicle images of the GL-UFD dataset, respectively. The global scene branch outputs the unnormalized global class-score vector $\mathbf{z}_g \in \mathbb{R}^6$, global feature vector $\mathbf{f}_g$, and global token sequence $\mathbf{T}_g$. The local vehicle-evidence

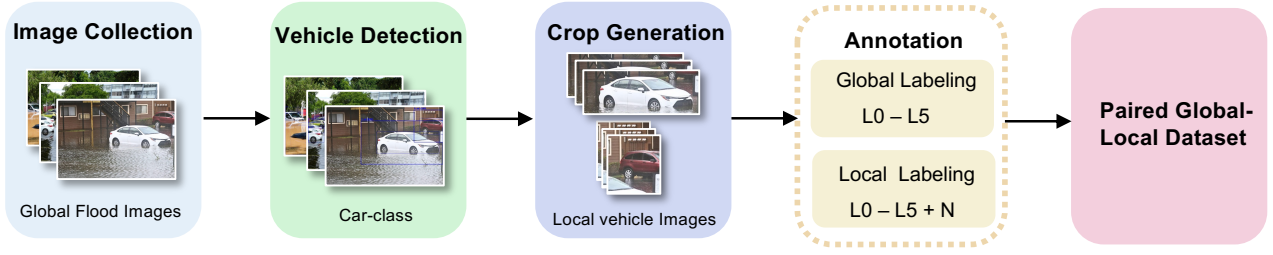


Fig. 2. The pipeline of global-local urban flood-depth (GL-UFD) dataset construction. Vehicles are detected by YOLOv11 vehicle detection [20], and then are cropped by expanding by 1.5 times around the detected bounding boxes. Flood depth is manually labeled at both the global and local levels.

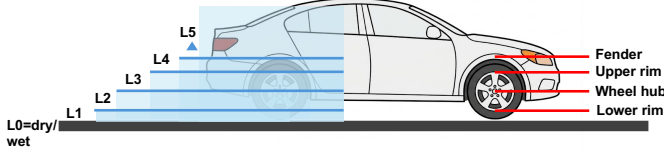


Fig. 3. Vehicle-referenced ordinal flood-depth labels. L0 denotes dry/wet road conditions, while L1–L5 are defined by waterline position relative to wheel lower rim, wheel hub, upper rim, fender, and above-fender depth.

branch extracts local feature vectors $\mathbf{f}_{c_i}$ and class probability vectors $\mathbf{p}_{c_i}$ for each detected vehicle. VE-SRM selects the most informative vehicle evidence and refines its feature using scene context. DRG-CM then produces the final prediction by applying bounded upward or downward ordinal corrections to $\mathbf{z}_g$, allowing local vehicle evidence to guide but not replace scene-level reasoning.

B. Vehicle-Evidence Selection and Refinement Module

Vehicle-evidence selection: The VE-SRM selects the most flood-informative local vehicle and refines its features using the global scene. For each detected vehicle $\mathbf{I}_{c_i}$, the local vehicle-evidence backbone outputs probabilities $p_{i,\ell}$ over local flood depth labels, where $\ell = 0, \dots, L-1$ is the ordinal depth, $L = 6$ for the six flood depth levels. Since the N-class is not assigned an ordinal depth, an expected-depth score s_i is computed from the probabilities of L0–L5:

$$s_i = \sum_{\ell=0}^{L-1} \ell * p_{i,\ell} \quad (1)$$

The vehicle with the highest expected-depth score s_{c^*} is then selected as the local vehicle evidence. This soft score is more stable than a hard predicted label because it uses the full ordinal probability distribution. If the selected vehicle (evidence) is indeterminate, occluded, or outside the flood-relevant region, its influence will be limited in the DRG-CM, so that the model can fall back toward the global scene anchor.

Vehicle-evidence refinement: Let $\mathbf{f}_{c^*}$ denote the selected local vehicle features, which encode local wheel and waterline evidence. This local evidence is then refined given the global scene context using the cross-attention mechanism. Specifically, the selected local features $\mathbf{f}_{c^*}$ are linearly projected

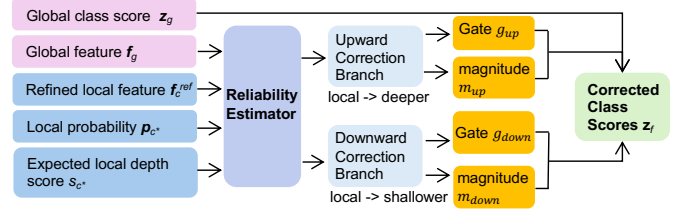


Fig. 4. An illustration of the directional reliability-gated correction module (DRG-CM). The reliability estimator separates upward and downward correction paths, allowing local vehicle evidence to adjust the prediction of the global scene only when the direction and magnitude are reliable.

into the query, while the global scene tokens $\mathbf{T}_g$ are linearly transformed into the key and value:

$$\text{Attmap}(\mathbf{Q}, \mathbf{K}) = \text{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}} \right) \quad (2)$$

where $\mathbf{Q} = \mathbf{W}_q \mathbf{f}_{c^*}$, $\mathbf{K} = \mathbf{W}_k \mathbf{T}_g$, and $\mathbf{V} = \mathbf{W}_v \mathbf{T}_g$ with learnable weight matrices $\mathbf{W}_q$, $\mathbf{W}_k$, and $\mathbf{W}_v$; d_k is the dimension of $\mathbf{K}$.

Then, the scene-conditioned vehicle-evidence features are calculated as

$$\mathbf{f}_c^{\text{ref}} = \text{Attmap}(\mathbf{Q}, \mathbf{K}) \cdot \mathbf{V} \quad (3)$$

This attention mechanism is consistent with the role of the local vehicle-evidence branch: the selected vehicle provides local waterline evidence, while the global scene helps determine whether that evidence is contextually reliable.

C. Directional Reliability-Gated Correction Module

As illustrated in Fig. 4, the DRG-CM decides whether the refined vehicle evidence should push the prediction of the global scene higher/lower or have little effect. This direction-aware correction is important because local vehicle evidence can play different roles in different scenes: it may correct global underestimation, be ignored when unrepresentative, or reduce global overestimation caused by misleading water regions, reflections, or non-road water bodies.

Let $\mathbf{h} = [\mathbf{f}_g; \mathbf{f}_c^{\text{ref}}; \mathbf{p}_{c^*}; s_{c^*}]$ denote the input to the reliability estimator. The correction module uses two lightweight multilayer perceptrons (MLPs): one MLP predicts directional reliability gates (g_{up} and $g_{\text{down}} \in (0, 1)$), and the other predicts correction magnitudes (m_{up} and $m_{\text{down}} \in (0, s_{\text{max}})$):

$$g_{\text{up}}, g_{\text{down}} = \sigma(\text{MLP}_g(\mathbf{h})), \quad (4)$$

$$m_{\text{up}}, m_{\text{down}} = s_{\text{max}} \sigma(\text{MLP}_m(\mathbf{h})), \quad (5)$$

where $\sigma(\cdot)$ is the sigmoid function. In our experiments, $s_{\text{max}} = 2.0$. Let $\mathbf{p}_g = \text{softmax}(\mathbf{z}_g)$ be the class probability vector of the global-scene-based prediction. Following Eq. 1, the expected global depth score is computed as $s_g = \sum_{\ell=0}^{L-1} \ell * p_{g,\ell}$, where $L = 6$ is the number of flood-depth levels. For each depth level ℓ , an ordinal direction template is computed from the global class scores as follows:

$$d_\ell(\mathbf{z}_g) = \frac{(\ell - s_g)/(L-1)}{\max_{j \in \{0, \dots, L-1\}} |(j - s_g)/(L-1)|}. \quad (6)$$

Thus, $\mathbf{d}(\mathbf{z}_g) \in [-1, 1]^L$. Adding this template increases evidence for classes deeper than the current global expectation, while subtracting it increases evidence for shallower classes. The corrected class-score vector is computed as

$$\mathbf{z}_f = \mathbf{z}_g + g_{\text{up}} m_{\text{up}} \mathbf{d}(\mathbf{z}_g) - g_{\text{down}} m_{\text{down}} \mathbf{d}(\mathbf{z}_g), \quad (7)$$

where $\mathbf{z}_f \in \mathbb{R}^6$ denotes the corrected unnormalized class scores for L0–L5. The final probabilities are $\mathbf{p}_f = \text{softmax}(\mathbf{z}_f)$, and the predicted flood-depth level is $\hat{y} = \arg \max_l p_{f,l}$.

D. Training Objective

Flood depth levels are ordered, so the global and local backbones are first trained with a cumulative-link ordinal objective [22]. For the global backbone, the input is the whole image $\mathbf{I}_g$; for the local backbone, the input is the i -th cropped vehicle image $\mathbf{I}_{c_i}$. The global and local backbones predict $\hat{r}_l^g = P(y^g > l \mid \mathbf{I}_g)$ and $\hat{r}_{i,l}^c = P(y_i^c > l \mid \mathbf{I}_{c_i})$, respectively. For either case, cumulative-link learning represents a label y by ordered binary targets [23]:

$$r_l = \mathbb{I}[y > l], \quad l = 0, \dots, L-2, \quad (8)$$

where $L = 6$ for L0–L5. The cumulative-link loss is

$$\mathcal{L}_{\text{cum}} = - \sum_{l=0}^{L-2} [r_l \log \hat{r}_l + (1 - r_l) \log(1 - \hat{r}_l)]. \quad (9)$$

This objective reflects the ordinal structure because deeper labels pass more ordered thresholds.

Both global and local backbones are frozen after pretraining. Then, VE-SRM and DRG-CM are trained with an overall loss function with six loss terms:

$$\mathcal{L} = \mathcal{L}_{\text{cum}_f} + \lambda_{\text{exp}} \mathcal{L}_{\text{exp}} + \lambda_{\text{bf}} \mathcal{L}_{\text{bf}} + \lambda_{\text{anch}} \mathcal{L}_{\text{anch}} + \lambda_{\text{rel}} \mathcal{L}_{\text{rel}} + \lambda_{\text{dir}} \mathcal{L}_{\text{dir}}. \quad (10)$$

$\mathcal{L}_{\text{cum}_f}$ is the cumulative-link classification loss of the final prediction:

$$\mathcal{L}_{\text{cum}_f} = - \sum_{l=0}^{L-2} [r_l \log \hat{r}_{f,l} + (1 - r_l) \log(1 - \hat{r}_{f,l})], \quad (11)$$

where $\hat{r}_{f,l} = \sum_{k=l+1}^{L-1} p_{f,k}$ is the probability that the final corrected prediction exceeds threshold l .

The expected-depth loss $\mathcal{L}_{\text{exp}}$ penalizes the deviation between the expected depth and the ground truth label and is defined as below:

$$\mathcal{L}_{\text{exp}} = \text{SmoothL1} \left(\frac{\sum_{\ell=0}^{L-1} \ell * p_{f,\ell}}{L-1}, \frac{y}{L-1} \right), \quad (12)$$

where $p_{f,\ell}$ is the final probability of level ℓ . SmoothL1 is a robust regression loss applied to the normalized expected depth and normalized ground-truth level.

The big-flip term $\mathcal{L}_{\text{bf}}$ penalizes probability mass assigned to classes more than two levels away from the ground truth (GT).

$$\mathcal{L}_{\text{bf}} = - \log \left(1 - \sum_{\ell: |\ell-y| \geq 3} p_{f,\ell} \right), \quad (13)$$

The anchor term $\mathcal{L}_{\text{anch}}$ keeps the corrected prediction close to the global-scene prediction as follows:

$$\mathcal{L}_{\text{anch}} = D_{\text{KL}}(\text{softmax}(\mathbf{z}_g) \parallel \text{softmax}(\mathbf{z}_f)), \quad (14)$$

where $D_{\text{KL}}(\cdot)$ is Kullback–Leibler divergence.

The relation term $\mathcal{L}_{\text{rel}}$ is a weighted cross-entropy loss that supervises whether $y_{c^*} < y$, $y_{c^*} = y$, or $y_{c^*} > y$, where y_{c^*} and y are the GT labels of the selected vehicle and the global-scene, respectively. The directional term $\mathcal{L}_{\text{dir}}$ supervises g_{up} and g_{down} : if the prediction of the selected vehicle is closer to y than the global prediction, the upward gate is encouraged when the selected vehicle prediction is deeper, and the downward gate is encouraged when it is shallower; otherwise, both gates are supervised toward 0.1, yielding weak vehicle correction. In the experiments, we set $\lambda_{\text{exp}} = 0.05$, $\lambda_{\text{bf}} = 0.1$, $\lambda_{\text{anch}} = 0.005$, $\lambda_{\text{rel}} = 0.1$, and $\lambda_{\text{dir}} = 0.2$, empirically.

V. EXPERIMENTS

A. Experimental Setup and Metrics

To evaluate the effectiveness of the proposed SF-VE fusion model, we compared it to two baseline models: the global-scene-based model and a late feature-level fusion model. Specifically, the global-scene-based baseline is the same as the global-scene branch used in the fusion models. The feature-level fusion baseline uses the same frozen global and local backbones, mean-pools valid local vehicle features, concatenates the pooled local feature with the global feature, and predicts the flood-depth label using a two-layer MLP classifier with a hidden dimension of 128, ReLU activation, dropout, and a six-class output layer. This baseline does not include vehicle-evidence selection, directional correction, or reliability gates.

Implementation Details: The global-scene backbone is fine-tuned with Swin-S pretrained on global images of GL-UFD dataset, with input size 448×448 , dropout 0.3, learning rate 2×10^{-5} , weight decay 1×10^{-3} , and the first two Swin stages frozen. The local vehicle backbone uses Swin-S with input size 224×224 , dropout 0.3, learning rate 2×10^{-5} , weight decay 1×10^{-3} , MixUp 0.1, and random erasing 0.25.

TABLE II
OVERALL QUANTITATIVE COMPARISON.

Model	Acc	Macro-F1	W-F1	MAE	BFR
Global-scene	0.6541	0.6418	0.6596	0.4261	0.0075
Late feature-level fusion	0.6817	0.6684	0.6878	0.3910	0.0100
SF-VE fusion	0.7143	0.7066	0.7203	0.3609	0.0075

The best validation checkpoints of these two backbones are then frozen and used by both fusion-based models.

The late feature-level fusion baseline uses the frozen global and local backbones, mean-pools valid vehicle features, concatenates them with the global feature, and predicts the flood-depth label using a two-layer MLP with hidden dimension 128, GELU activation, dropout 0.5, weight decay 1×10^{-2} , and learning rate 2×10^{-5} . The proposed SF-VE model uses hidden dimension 64, attention dimension 64, four attention heads, dropout 0.6, weight decay 2×10^{-2} , and fusion-head learning rate 1×10^{-5} . The relatively high dropout is used to reduce overfitting because the fusion head is trained on a 2,660-image paired dataset while receiving strong frozen backbone features. Both fusion models are trained with AdamW, physical batch size 8, gradient accumulation over 4 steps, and validation-based checkpoint selection for up to 100 epochs.

Evaluation Metrics: Because flood-depth estimation is both a classification and ordinal prediction problem, accuracy, macro-F1, weighted-F1, mean absolute error (MAE), and big-flip rate (BFR) are reported for all three models. BFR is the fraction of predictions with an absolute error of at least three levels, which is important for safety because these errors can change a scene from shallow to severe flooding or the reverse [24].

B. Overall Results

Table II summarizes a quantitative comparison of the three models. The global-scene-based model provides a strong scene-level baseline. The late feature-level fusion model improves over the global-scene-based baseline, indicating that local vehicles provide complementary information. However, its higher BFR suggests that naive fusion may amplify misleading vehicle evidence. The proposed SF-VE fusion model achieves the best performance among all metrics. Compared with the late feature-level fusion model, SF-VE improves accuracy by 3.26%, macro-F1 by 3.82%, and weighted-F1 by 3.25%, respectively, and reduces MAE from 0.3910 to 0.3609, and BFR from 0.0100 to 0.0075.

C. Per-Class Performance

Table III reports per-class F1 scores, and Fig. 5 further compares the confusion matrices of the feature-level fusion baseline and the SF-VE fusion model. Compared with the global-scene-based baseline and late feature-level fusion baseline, the proposed SF-VE model improves F1 scores for all six flood-depth levels. The improvements are more pronounced for L1 and L4.

The confusion matrices show that the proposed SF-VE fusion model reduces errors between adjacent classes while

TABLE III
PERFORMANCE COMPARISON IN TERMS OF PER-CLASS F1 SCORES.

Model	L0	L1	L2	L3	L4	L5
Global-scene	0.706	0.622	0.627	0.493	0.564	0.839
Feature-level fusion	0.727	0.644	0.670	0.534	0.582	0.853
SF-VE fusion	0.765	0.712	0.694	0.559	0.649	0.860

TABLE IV
ABLATION STUDY OF SF-VE CORRECTION VARIANTS IN TERMS OF PER-CLASS F1 SCORES.

Variant	L0	L1	L2	L3	L4	L5
w/o Gates	0.686	0.646	0.606	0.554	0.587	0.831
w/ Single gate	0.765	0.708	0.694	0.535	0.632	0.854
w/ Directional gates	0.765	0.712	0.694	0.559	0.649	0.860

preserving a strong prediction performance for L5. Strong improvements occur at operationally important middle levels, where small differences in the waterline around the wheel hub, upper rim, and fender can change the safety interpretation of a scene. These results support the main design choice: local vehicle evidence is useful when controlled by scene-aware reliability rather than simply concatenated with the global feature.

D. Ablation Study

The ablation study focuses on how local vehicle evidence is employed to influence the prediction of the global scene. Using the same frozen global and local backbone fusion setting, three variants of the proposed DRG-CM were evaluated to test whether bounded and direction-aware correction is important for a reliable use of vehicle-based evidence.

The three ablation variants differ in how the correction term is controlled. The *w/o-gates variant* removes reliability gates by setting the correction gates to one:

$$\mathbf{z}_f = \mathbf{z}_g + m_{\text{up}} \mathbf{d}(\mathbf{z}_g) - m_{\text{down}} \mathbf{d}(\mathbf{z}_g). \quad (15)$$

The single-gate variant uses one scalar gate $g \in (0, 1)$ for the whole correction:

$$\mathbf{z}_f = \mathbf{z}_g + g [m_{\text{up}} \mathbf{d}(\mathbf{z}_g) - m_{\text{down}} \mathbf{d}(\mathbf{z}_g)]. \quad (16)$$

The proposed directional-gate variant uses separate upward and downward gates:

$$\mathbf{z}_f = \mathbf{z}_g + g_{\text{up}} m_{\text{up}} \mathbf{d}(\mathbf{z}_g) - g_{\text{down}} m_{\text{down}} \mathbf{d}(\mathbf{z}_g). \quad (17)$$

Table IV shows that uncontrolled local correction can be harmful. The no-gate variant performs worse than the global-scene baseline on several classes because selected vehicle evidence is not always representative of the scene-level flood condition. For example, a detected vehicle may be located on an elevated dry road, outside the deepest flooded region, or near a non-road water body. Without reliability gates, such local evidence can perturb and degrade a correct scene-level estimate. This supports the scene-first design: local vehicle evidence should correct the global prediction only when it

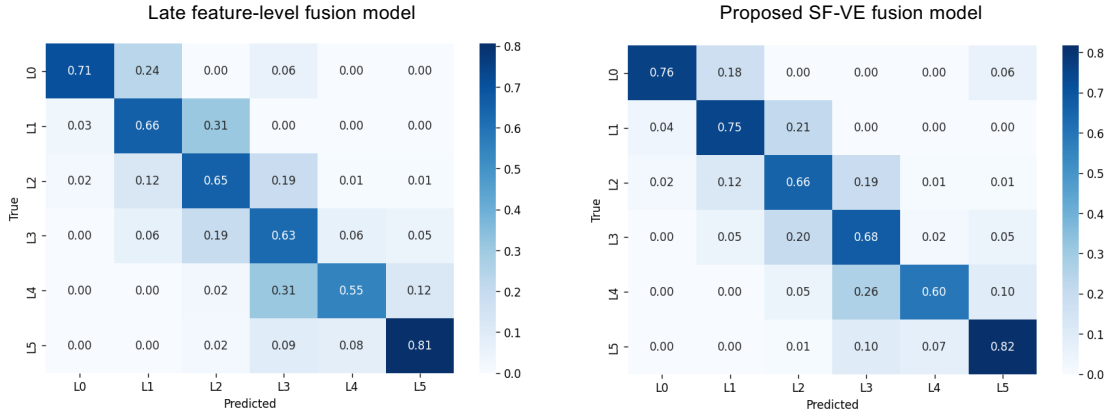


Fig. 5. Confusion matrices of the late feature-level fusion and the proposed SF-VE fusion. Values are row-normalized by ground-truth class.

is reliable. Adding a single gate improves most classes, confirming the importance of reliability control. However, the single gate performs worse than the no-gate variant for L3, suggesting that a direction-agnostic reliability score can over-suppress useful middle-depth evidence. The directional version preserves the gains of reliability control while improving L1, L3, L4, and L5, supporting the use of vehicle evidence as direction-aware correction rather than as generic additional features.

E. Qualitative Visualization and Error Analysis

To examine how different models use scene and vehicle-based evidence, Grad-CAM [25] is employed to visualize class-discriminative regions for the predicted flood-depth label. Fig. 6 shows that the proposed SF-VE model uses local vehicle evidence conditionally rather than treating it as a direct substitute for scene-level reasoning.

In Case A, the global-scene-based model underestimates flood depth, while the selected vehicle provides clearer local waterline evidence. The proposed SF-VE model successfully uses this evidence to correct the prediction upward as indicated in the highlighted regions around the front part of the SUV, which is submerged in deeper water.

In Case B, all models predict the correct severe flood level, but the role of local evidence is different. The selected vehicle is small and elevated, while the global scene contains strong evidence of severe flooding. The proposed SF-VE heatmap remains concentrated on the broader flooded road region rather than relying only on the selected vehicle. This indicates that the correction module does not force local vehicle evidence to change the prediction when scene-level evidence is already reliable.

Case C shows an example of the opposite direction. The global-scene-based model and late feature-level fusion baseline overestimate the flood level, while the selected vehicle evidence supports a shallower water depth. The proposed model reduces the final prediction accordingly, as shown in the highlighted regions around the selected vehicle.

Case D shows an example of failures: a river or canal region visually resembles deep roadway flooding, causing the global-scene-based model to predict an overly severe level. Since the selected local vehicle evidence in the dark blue region on the heat map is considered not informative, the proposed SF-VE model discards the local evidence and preserves the scene-level prediction. This suggests that future work should incorporate flood-region or road-water awareness to better distinguish roadway flooding from natural or non-road water bodies.

VI. CONCLUSION

This paper presented a scene-first vehicle-evidenced (SF-VE) fusion model and a paired global-local urban flood-depth (GL-UFD) dataset for ordinal urban flood-depth estimation. The dataset links global urban flood scenes with cropped vehicle images and uses a vehicle-based L0–L5 rule developed in this work for road-safety-oriented interpretation. The proposed SF-VE fusion model keeps the global image as the scene anchor, uses VE-SRM to select and refine vehicle evidence with scene context, and employs DRG-CM to produce bounded upward or downward ordinal corrections.

Experiments show that local vehicles provide useful conditional evidence, but are not a substitute for scene-level reasoning. Compared with the late feature-level fusion, the proposed SF-VE model improves all metrics including accuracy, macro-F1, weighted-F1, MAE, and per-class F1 score across all flood depth levels, while keeping big-flip errors low. The ablation study further shows that reliability control and directional correction are important for stable global-local fusion.

Failures occur when the flood-relevant vehicle is missed, the deepest flooded region appears far in the background, or rivers and canals resemble roadway flooding. Future work will explore flood-region-aware supervision, weak labels for road flooding versus non-road water bodies, and broader reference cues such as pedestrians and traffic infrastructure.

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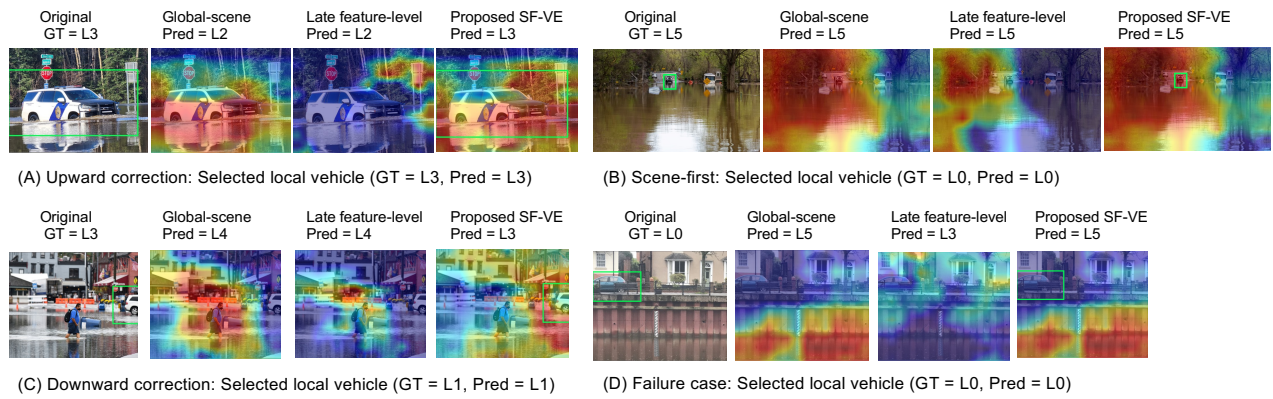


Fig. 6. Qualitative comparison of the three models using heat maps generated by Grad-CAM [25]. The four cases illustrate upward correction, scene-first preservation when the selected vehicle is unrepresentative, downward correction, and a failure case due to river/canal ambiguity, respectively. Each case shows the original image with the selected vehicle evidence marked by a green box, followed by heatmaps from the global-scene-based baseline, late feature-level fusion baseline, and proposed SF-VE model, respectively. Heatmaps are mapped back to the original image region, where warmer colors indicate stronger class-discriminative evidence for the predicted flood-depth level.

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